

Day 02

Classification with KNN and `scikit-learn`

MSU REU Machine Learning Short Course

The Task

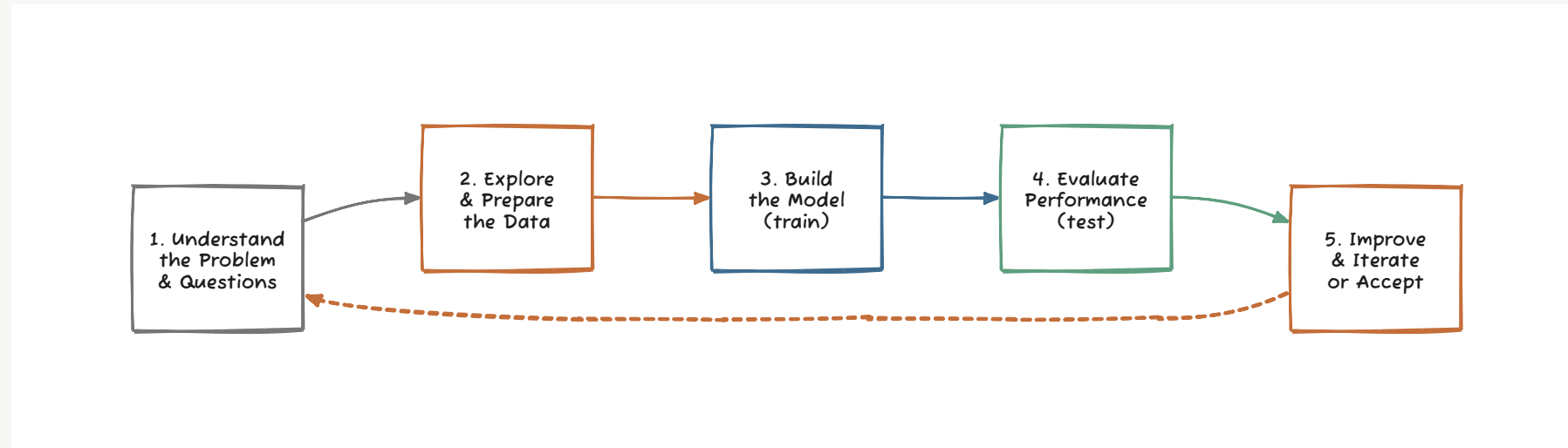
Given photometric measurements of an object, predict whether it is a **STAR**, **GALAXY**, or **QSO**.

- Yesterday: we **explored** the data
- Today: we **model** it

The model learns a **decision boundary** — a rule in feature space that separates classes.

The ML Pipeline

Every model you build this week follows this same structure.



Learn it once. It applies to classification **and** regression.

Train / Test Split and Scaling

Why split? You can't evaluate yourself on the data you learned from.

```
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler

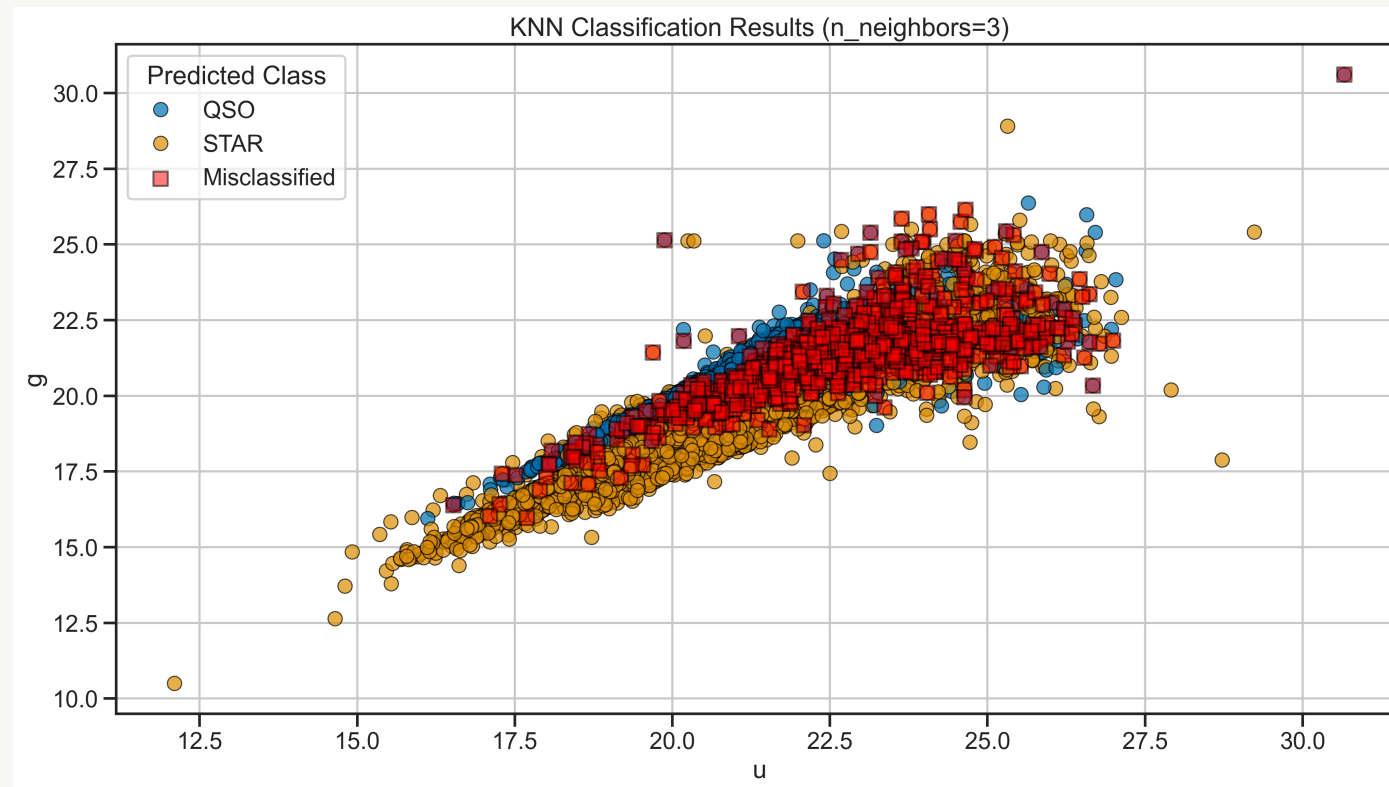
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)      # same scale as train
```

Why scale? KNN uses distance. A feature with range 0–1000 dominates one with range 0–1.

KNN — You are your neighbors

To classify a new point, find the **k** closest training points and take a majority vote.



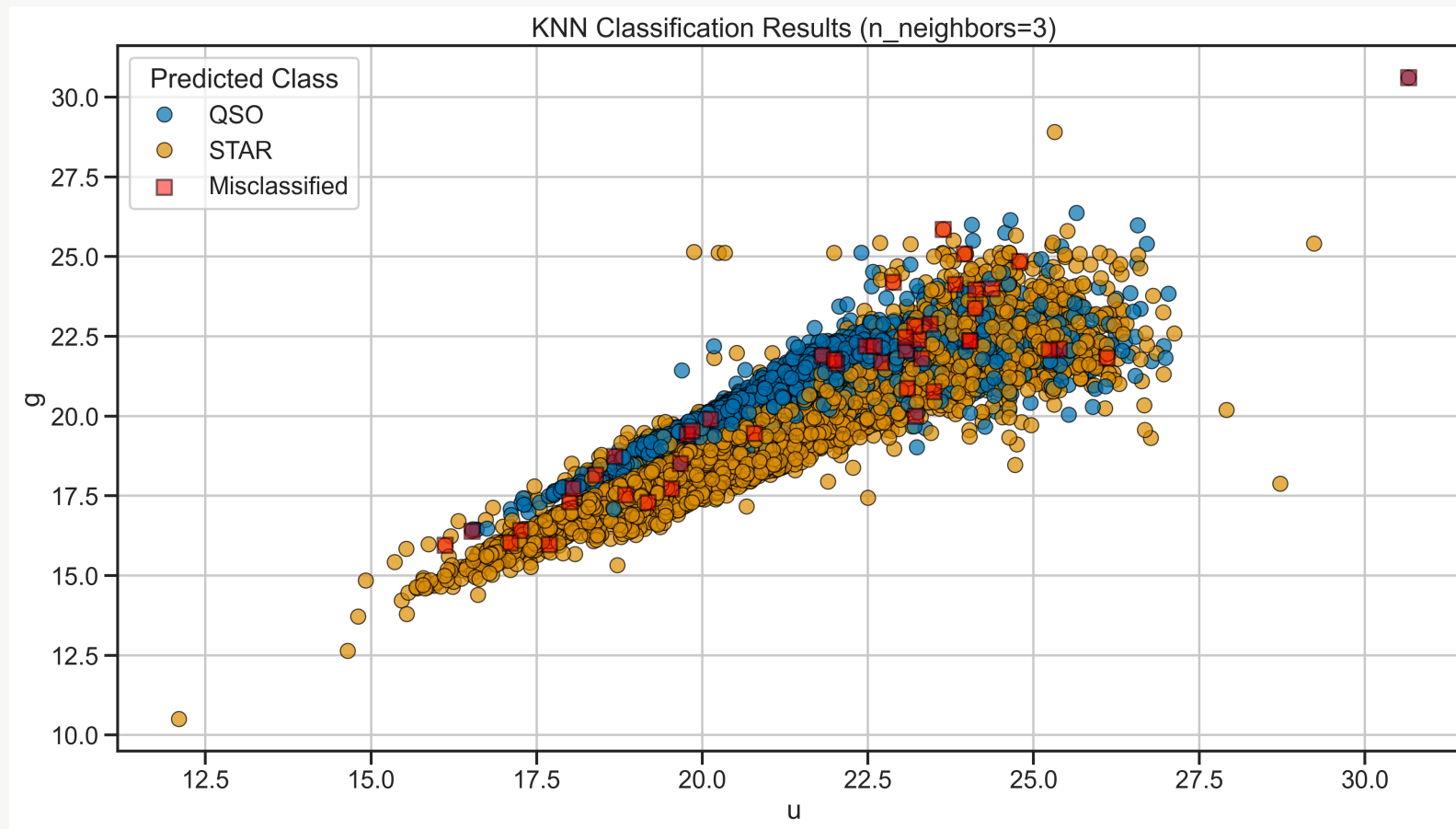
Evaluation — The Confusion Matrix

Accuracy alone hides the interesting failures.

True Positive (TP) Said YES · Was YES ✓	False Positive (FP) Said YES · Was NO ✗
False Negative (FN) Said NO · Was YES ✗	True Negative (TN) Said NO · Was NO ✓

Redshift changes everything

Without redshift: STAR vs QSO is hard. With it: near perfect.



Today's Activity

Work through **Activity 02: Classification with scikit-learn**

1. Build a 2-class KNN (STAR vs QSO) without redshift
2. Tune **k** and observe the effect on performance
3. Add redshift — compare the confusion matrices
4. Extend to 3-class (STAR, GALAXY, QSO)

Notes: [Methods & Validation](#) · [Support Vector Machines](#)